复合乘积：展开后再按分量重组

$$[\nabla(\mathbf{A}\cdot\mathbf{B})]_i = \partial_i(A_jB_j) = (\partial_iA_j)B_j + A_j(\partial_iB_j)$$

$$\nabla\cdot(\mathbf{A}\times\mathbf{B}) = \partial_i(\varepsilon_{ijk}A_jB_k) = \varepsilon_{ijk}(\partial_iA_j)B_k + \varepsilon_{ijk}A_j(\partial_iB_k)$$

$$\nabla\cdot(\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot(\nabla\times\mathbf{A}) - \mathbf{A}\cdot(\nabla\times\mathbf{B})$$

负号来自交替指标的交换。